

Essential oils and their bioactive molecules in healthcare

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11.1 Introduction

The word “essential oil” (EO) was supposedly first coined by Paracelsus von Hohenheim in the sixteenth century from the term *Quinta essentia*, meaning effective component of a drug [1]. EOs are natural, aromatic, and volatile liquid produced by plants as a by-product of their secondary metabolism. More than 17,500 plant species, mostly from angiosperm families are known to produce EOs [2]. Depending upon the culture and traditions of different countries, EOs have been widely used worldwide in foods, cosmetics, perfumeries, feeds, beverages, aromatherapies, and pharmaceuticals [3]. Egyptians pioneered the utilization of EOs (as early as 4500 BCE) from aromatic plants for both prevention and cure of many diseases and their therapeutic knowledge indeed inspired Greeks too, during which Hippocrates employed EOs in aromatherapy to improve their quality of life [4,5]. In due course, Romans started adopting Greek medicine, particularly after knowing the impressive power of the fragrance (essence) in the aromatherapy. Aromatherapy

(*aroma*—smell or fragrance; *therapy*—to heal) uses aromatic substance like EOs to heal one’s body, mind, and soul [3]. Chinese and Indians have historical records of more than 700 aromatic herbs used for healing purposes [6]. EOs are becoming very popular in Chinese traditional medicines, where EOs are widely applied in the aromatherapy and massage therapy [7]. Steam bath therapy infused with oils from jasmine (*Jasminum* sp.), ylang-ylang (*Cananga odorata*), and lavender (*Lavandula hybrida*) are known to stimulate the central nervous system and help in relaxation [8,9]. In Bhutan, EOs are used for lighting purposes and as ointments for spiritual healing (*ngag-dmar*).

EOs affect our body either by absorption through skin or binding to olfactory receptors which are connected to the brain. EOs are complex mixture of different components, making a strong cocktail of biomolecules which can cause severe side effects if it is not diluted using carrier oils before use. Thus, oils from sunflower, soybean, avocado, sweet almond, and sesame seed are usually used as carrier oil at a concentration of 10%–25% to obtain optimum benefits.

Similarly perfumes contain only 25% fragrance, while the rest 75% is either alcohol or other carrier oils [10] which releases scent/ or aroma as the alcohol evaporates. EOs are widely distributed in nature, and until now, more than 3000 EOs are studied from plants, of which 300 are available in the fragrance and flavor markets [2]. The global market value for EO is anticipated to reach over USD 15 billion by 2025 [11]. Some of the most commonly traded EOs and the countries producing them are given in Fig. 11.1.

This chapter presents EOs, their chemistry, sources, methods of extraction, and their scope for the development of therapeutic drugs. We have also discussed the commercial applications of EOs briefly highlighting the global trade markets and potential risks associated with EOs applications.

11.2 Sources of essential oils

EOs are widely distributed in nature. There are a few EOs from animal sources which are

used as ingredients in perfumes, such as, Civet oil from civets' odorous sacs (*Paradoxurus hermaphroditus*), castoreum from the North American beaver's (*Castor canadensis*) odorous sacs, beeswax absolute distilled from the honeybee's honeycomb, and ambergris oil from Sperm Whales (*Physeter macrocephalus*) [10]. However, plants are considered as the major source of EOs. EOs are secondary metabolites in the plants, and they are either defensive in function or assist in the reproduction by producing scents to attract pollinators. More than 1700 individual aroma compounds are identified from 990 plus plant taxa belonging to 90 families and 38 orders [12]. Approximately 6% of total 295,383 angiosperms identified so far are known to yield EOs, and most of the EO-producing species belong to *Lamiaceae*, *Rutaceae*, *Myrtaceae*, *Zingiberaceae*, and *Asteraceae* families [2]. Almost all parts of the plant produce EO (Table 11.1), but their yield differs among different parts. In plants, EOs are usually stored in their cavities, secretory cells, canals,



FIGURE 11.1 Major essential oil producing countries and common names of plant sources.

TABLE 11.1 Representative essential oil producing plants, extraction methods, major oil constituents and their reported biological activities.

Plants and family	Part(s) used	Method of extraction	Major components of essential oil	Pharmacological properties
<i>Allium sativum</i> (Amaryllidaceae)	Bulbs	HD-CT	Diallyl trisulfide, diallyl disulfide, allyl methyl trisulfide, diallyl sulfide, and diallyl tetrasulfide, of total 22 compounds identified.	Anti-inflammatory [13]
<i>Citrus sinensis</i> (Rutaceae)	Fruit peel	CP	Limonene, limonene 1,2-epoxide, cis- <i>p</i> -mentha-2,8-dien-1-ol and (<i>E</i>)-carveol, of total 24 compounds identified.	Anthelmintic [14,15]
<i>Commiphora pyracanthoides</i> (Burseraceae)	Sap from stem bark	HD	2-Cyclohexen-1-one and 4-ethynyl-4-hydroxy-3,5,5-trimethyl, of total 76 compounds identified.	Anticancer [16]
<i>Croton tiglium</i> (Euphorbiaceae)	Fruits	SCE	17-Octadecynoic acid, tetradecanoic acid, 17-Octadecynoic acid methyl ester, <i>n</i> -hexadecanoic acid, <i>n</i> -decanoic acid, linoleic acid ethyl ester, and i-Propyl 9-octadecenoate, of total 28 compounds identified.	Anticancer [17]
<i>C. citratus</i> (Poaceae)	Leaves	SD	Geranial, neral, β -myrcene, geranyl acetate, of total 23 compounds identified.	Antiplasmodial [18,19]
<i>Fortunella margarita</i> (Rutaceae)	Fruit peel	HD	Limonene, myrcene, and δ -germacrene, of total 45 compounds identified.	Cardioprotective [20,21]
<i>Hedychium coronarium</i> (Zingiberaceae)	Rhizome	HD-CT	β -pinene, eucalyptol, linalool, coronarin-E, α -pinene, <i>p</i> -cymene, γ -terpinene, and 10-epi- γ -eudesmol, of total 44 compounds identified.	Antiparasitic [22]
<i>Juniperus communis</i> (Cupressaceae)	Leaves and inflorescence	CCE	Limonene and α -pinene, of total 79 compounds identified.	Anticancer [23,24]
<i>Lavandula hybrida</i> (Lamiaceae)	Flowers	THD, Salt-HD, Micelle-HD, Enzyme-HD, SFME, United States-SD, WS-HD	Linalool and linalyl acetate out of 47 compounds identified.	Antifungal [25,26]
<i>L. sibiricus</i> (Lamiaceae)	Normal roots	HD-CT	β -Selinene, selina-4,7-diene, (<i>E</i>)- β -caryophyllene, myli-4(15)-ene, and guaia-1(10),11-diene, of total 70 compounds identified.	Antifungal [27]
<i>Mentha spicata</i> (Lamiaceae)	Leaves and stems	HD	Carvone, limonene, and β -pinene of total 12 compounds identified.	Antiflatulence [21,28]
<i>Ocimum americanum</i> (Lamiaceae)	Leaves	HD-CT	1,8-Cineol, camphor, α -pinene, and <i>trans</i> - α -bergamotene, of total 44 compounds identified.	Anesthetic [29,30]
<i>Pinus</i> spp. (Pinaceae)	Needles	HD-CT	α -Pinene, bornyl acetate, β -caryophyllene, α -guaiene, and germacrene D, of total 72 compounds from six species altogether.	Antibacterial [31,32]

(Continued)

TABLE 11.1 (Continued)

Plants and family	Part(s) used	Method of extraction	Major components of essential oil	Pharmacological properties
<i>Salvia lavandulifolia</i> (Lamiaceae)	Aerial	HD-CT	Camphor, 1,8-cineole, camphene, and α -pinene, of total 40 compounds identified.	Antioxidant [33,34]
<i>Syzygium aromaticum</i> (Myrtaceae)	Whole	SD	Betulinic acid and triterpenes.	Antiparasitic [35,36]
<i>Thymus bovei</i> (Lamiaceae)	Aerial	MSD	<i>trans</i> -Geraniol, α -citral, and β -citral, of total 70 compounds identified.	Antimicrobial, anti-oxidant, and anthelmintic [37]
<i>Trachyspermum ammi</i> (Apiaceae)	Seed	HD-CT	γ -Terpinene, thymol, and <i>p</i> -cymene, of total 23 compounds identified.	Antiviral [38,39]

Abbreviations: CCE, Conventional cleverger type extraction; CP, cold press extraction; *Enzyme-HD*, enzyme assisted hydro-distillation; HD, hydro-distillation; HD-CT, hydro-distillation assisted by cleverger apparatus; MSD, microwave-assisted steam distillation; *Salt-HD*, salt assisted hydro-distillation; SCE, supercritical CO₂ extraction; SD, steam distillation; SFME, solvent-free microwave extraction; THD, turbo hydro-distillation; *United States-SD*, ultrasound assisted hydro-distillation; *WS-HD*, subcritical water-based hydro-distillation.

epidermic cells, and glandular trichomes [40]. The EO concentration and composition is not the same across all plant species. Both yield and composition vary between genera as well as between species of the same genus. Such variation is attributed to multiple factors including ontogenic, genetic, and morphogenetic, geographical locations, maturity, post harvest drying, storage conditions, and plant part(s) used for extraction [41,42]. For examples, rosemary, eucalyptus, and lemongrass synthesize EOs in leaves and stems; almonds and celery in their seeds; Juniper and allspice in berries; cinnamon in the bark; citrus and grapefruits in fruit peel or rind; ginger in rhizomes; and chamomile, clove, jasmine, lavender, and rose in their flowers [43].

11.3 Chemistry of essential oils

EOs are a highly complex mixture of volatile lipophilic compounds with molecular weight less than 300, and they are mostly concentrated in flowers, seeds, leaves, rhizomes, and barks of plants [44]. Chemically, EOs contains a mixture of

complex and multifunctional compounds [45]. EOs based on their compositions are broadly classified into two categories, oxygenated compounds and hydrocarbons [40] (Fig. 11.2). Oxygenated constituents of EO are mostly alcohols (e.g., geraniol, α -bisabolol), aldehydes (e.g., citronellal, sinensal), esters (e.g., γ -terpinyl acetate, cedryl acetate), ketones (e.g., menthone, *p*-vetivone), oxides, phenols (e.g., thymol), and other aromatic and sulphur-containing groups [40,45,46]. Hydrocarbon constituents are characterized by a specific chemical group, terpenes [40]. Hydrocarbon constituents of EO are named based on the presence of varying numbers of isoprene units (C₅), such as monoterpenes (C₁₀), sesquiterpenes (C₁₅), and diterpenes (C₂₀), among which monoterpenes are major constituents (approximately 90%) of EOs [40]. Isoprenoids are synthesized from two isoprene units, isopentyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP) [47]. IPP and DMAPP precursors are formed via six steps mevalonate pathway (MVA) using acetyl-CoA and the non-MVP through condensation of pyruvate and glyceraldehyde-3-phosphate (G3P) with the help of seven enzymes [48]. EOs also contain

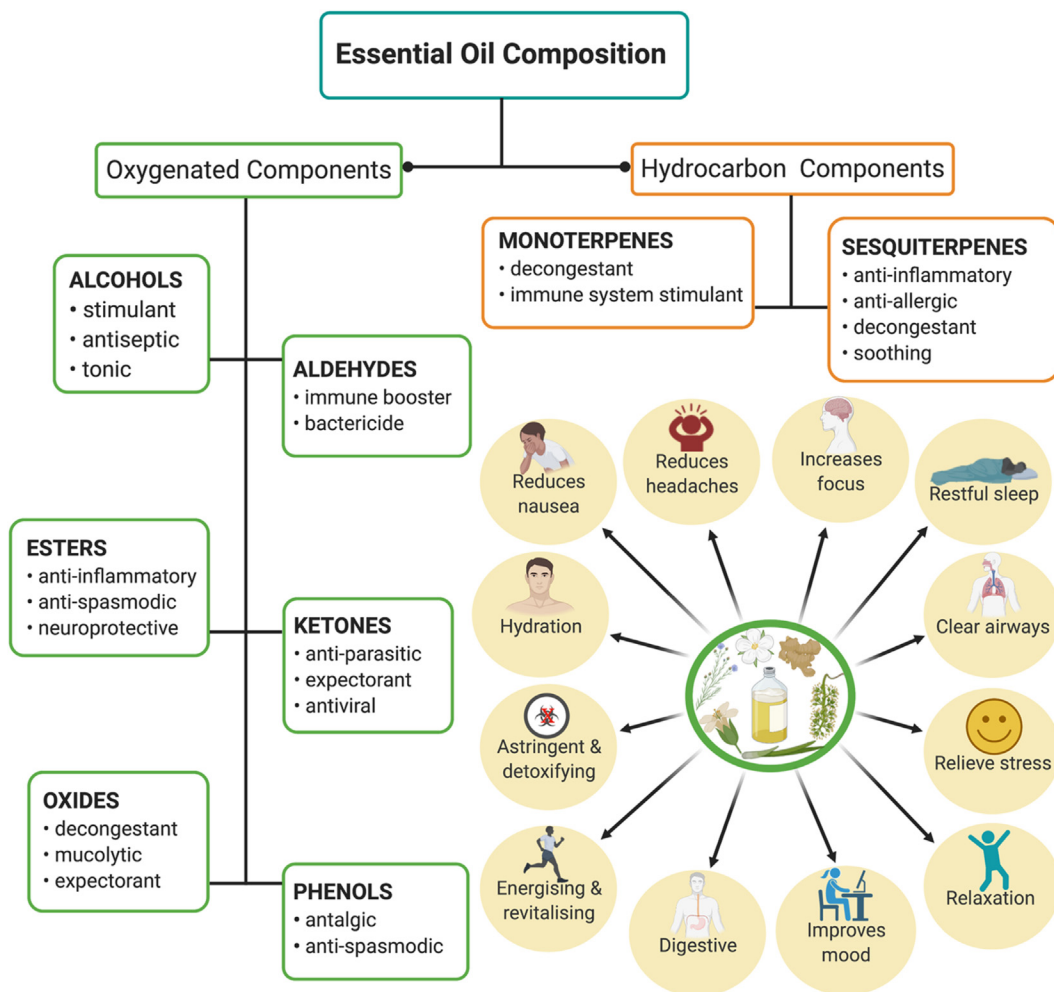


FIGURE 11.2 Major chemical components of essential oils and their general health benefits.

phenylpropanoid compounds derived from the shikimic acid pathway [40]. The EOs composition profile is influenced by the extraction techniques applied, the origin of the raw materials (plants), habitat soil profile, parts used, the age of the plants, and collection season [49]. However, there is no significant variation in the composition of EOs obtained from the plant parts grown under similar conditions [4]. Oil composition is also influenced by the parts used and harvesting season. Among the EOs obtained from different parts

of the same plant also varies, and it is attributed to their functional difference. For example, EO from *Cymbopogon martinii* (Roxb.) Wats. var. *motia* Burk. leaf contains geraniol as a major compound, while (*E*)- β -ocimene and scent compound geranyl acetate were present as major constituents in the oil from inflorescence [50]. Considering the harvest time, the EO yield from *Mentha longifolia* leaves is maximum during Spring. Moreover, there was also a slight variation in the oil composition characterized by pulegone, 1,8-cineole, and

L-menthone as major components in winter oil, and pulegone, oleic, and palmitic acids in the spring oil [51].

11.4 Methods of extraction and analysis of essential oils

Several methods and techniques are used for obtaining EOs from natural resources (Table 11.1). EOs from citrus plants are easily obtained by expression (squeezed) from their fruit peels. Hydro-distillation or steam distillation is widely used to isolate EOs from the aromatic plants as it allows easy recovery of oils from water due to their density differences [25]. The EO extraction techniques have advanced considerably from the conventional hydro-distillation method to more efficient and innovative methods, such as ultrasound-assisted extraction, microwave extraction, pressurized liquid extraction, instant controlled pressure drop technology, supercritical fluid extraction (SFE), and using alternative solvents (α -pinene, 2-methyl tetrahydrofuran, ethyl acetate, ethyl lactate, butanol, isopropanol, ethanol, dimethyl carbonate) to ensure minimal loss of oil components from heating (Table 11.1). However, not all these techniques or methods gives 100% yield, and the efficiency of methods relies upon multiple factors (e.g., the plant materials chosen for the EO extraction). For instance, in the extraction method using alternative solvents, the 2-methyl tetrahydrofuran solvent was promising for plants like *Ribes nigrum* L. [52]. The SFE technique is a better option for oil extraction from *Rosemarinus officinalis* [53], *Origanum vulgare*, *Salvia officinalis*, and *Thymus zygis* [54] and even in case of *Citrus* species like *C. sinensis* [52]. For some plants, a type of technique used for extraction does not cause variations in the oil yield. For instance, the yield of limonene (principal component) in orange peel (*C. sinensis*) oil was almost equivalent for both sono-Clevenger

(SC) (71.2%) and conventional Clevenger (CC) (71.5%) extraction processes, except for their extraction time difference (SC method was faster than CC process) [55].

In general, the choice of the right extraction method is very critical for the best yield of EOs. For solvent-assisted extraction methods, organic and nontoxic solvents are preferable to toxic petroleum-based solvents. A “Direct Green” extraction technique is one of such techniques which could avoid the loss of volatile components from the EOs. This technique use vegetable oils instead of petroleum-based solvents, such as dichloromethane [56]. Some plants do not show significant variations in their EO yield despite the nature of solvents used. The Lavender herb is one such example, and oil yield with 10 different extraction methods showed almost similar chemotype composition with 47 major compounds detected in all EOs [25]. For obtaining the best quality and maximum oil yield, hydro-distillation or steam distillation assisted by Clevenger-type apparatus is widely used for the extraction of EOs. The typical steam or hydro-distillation method of EO extraction is shown in Fig. 11.3A.

For the analysis of EO compositions, the gas chromatography (GC) is preferred over the liquid chromatography (LC) such as high-performance liquid chromatography (HPLC), as the later is not suitable for EO analysis due to high volatility and low UV absorptivity of most of its volatile constituents [57]. For the identification of known compounds from EOs, GC coupled with mass spectrometry (GCMS) is widely used. In GC, the retention time (Rt) is the main criterion for the identification of peaks. However, without additional information about compounds, it is nearly impossible to confirm the identity of peaks generated by GC with retention time alone, since compounds with similar volatility may elute at the same time [58]. Thus, MS is used to generate additional structural information of the compounds. The GC-MS has enabled rapid identification of EO

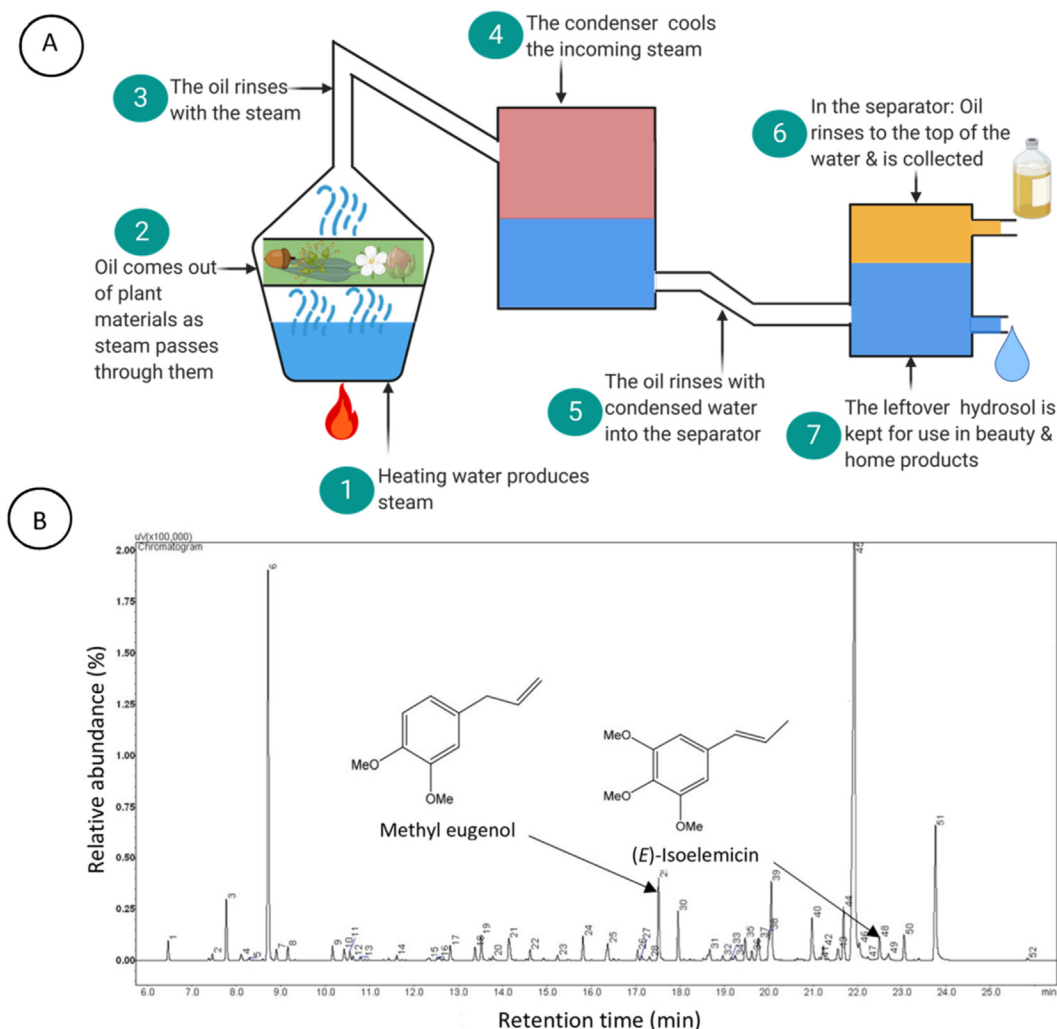


FIGURE 11.3 Essential oil extraction and analysis: (A) a typical hydro-distillation method of essential oil extraction; (B) An example of gas chromatography – flame ionization detector spectrum: Peaks showing the structure of isolated compounds from essential oil of Bhutanese medicinal plant *Pleurospermum amabile*. Source: Photo. P. Wangchuk.

compositions through comparison of ion fragmentation patterns with various MS library or online databases such as National Institute of Standards and Technology database (NIST14, which is available with many GC-MS instruments).

Though GC-MS, in general, is a best technique to generate compound identity, sometimes it is tricky, when different compounds yield similar

mass spectra. For example, citronellyl acetate often matches with 2,6-dimethyl-2,6-octadiene, and linalool with linalyl anthranilate, and for several cyclic *cis/trans* isomers and few sesquiterpenes too. Based on these examples, the mass spectrum alone is inadequate, and there are possibilities of misidentification; thus the incorporation of retention index into GC-MS is

recommended. Considering additional factors such as the limit of detection, the limit of quantification, and relative standard deviations (RSDs) may enhance the accuracy of compound identification and data reproducibility. To achieve this, one of the best available options is to couple a flame ionization detector (FID) with GC as it gives lower RSDs compared to GC-MS alone [59]. It can detect almost all organic compounds and has resistance toward even the slightest fluctuations of the gas flow, and its response is predictable, obeying the equal carbon response rule [60]. Thus GC-FID is the preferred technique for analyzing lipids. For the identification of unknown compounds isolated from EOs, a cross-sectional platform of GC-MS assisted by nuclear magnetic resonance (NMR) spectroscopy is applied, but it is expensive. A typical GC-FID spectrum is shown in Fig. 11.3B.

11.5 Biological activities of essential oils

EOs, owing to their diverse natural protective functions, humans have exploited EOs as antimicrobials, anticancer, antioxidants, antiparasitic, and antiinflammatory agents. Currently, about 3000 EOs are reported, of which about 300 are commercially available in the flavor and fragrance markets [2]. Most of the studies are conducted with whole crude EOs. The bioactive properties of EOs are attributed to the presence of a single or few bioactive constituents in the oil [61]. Some EOs tend to exhibit unique synergistic pharmacological effects when used in combination with other EOs [62]. For instance, compounds derived from EOs, such as carvacrol, γ -terpinene, and *p*-cymene confer better antifungal activity when used as a cocktail than as an individual compound [63]. EOs and their components possess broad-spectrum biological activities ranging from antimicrobial to anticancer activities, which are discussed in the subsequent sections. Studies have shown that

the biological properties of EOs are influenced by multiple factors, including plant parts used for EO extraction, maturity, morphogenetic factors, and environmental factors, including harvesting season. For example, *Salvia africana-lutea* EO collected in the winter tend to exhibit better antioxidant activity than those samples collected during other seasons [64]. Some common examples of bioactive molecules isolated and identified from various EOs are shown with their chemical structures in Fig. 11.4.

11.5.1 Anti-inflammatory biomolecules

EOs (crude as well as their components) have strong anti-inflammatory, antinociceptive, and immunomodulatory properties. Anti-inflammatory EOs either reduce edema, inflammatory cytokines (e.g., tumor necrosis factor alpha, TNF- α ; interleukin-1 beta, IL-1 β), inducible nitric oxide (iNO) production or reduces the expressions of prostaglandin E₂ and cyclooxygenase 2 (COX-2). There are many studies conducted on anti-inflammatory activities of whole EO and their components. Lavender EO is one of the oils widely studied for such properties [65]. EOs from Roman chamomile (*Anthemis nobilis*), German chamomile (*Matricaria recutita*), nutmeg (*Myristica fragrans*), clove (*Syzygium aromaticum*), cinnamon (*Cinnamomum zeylanicum*) have a considerable anti-inflammatory effect. EOs are easily absorbed into and through the skin, and this property makes them effective against topical inflammations. At times, applying EOs topically for topical inflammations is regarded with skepticism, but their components are widely used to enhance the delivery of oral as well as topical drugs [66]. For example, anti-inflammatory drug indomethacin is well absorbed by our body with the help of cardamom oil [67].

Some EO components, such as dehydrocostus lactone (from *Saussurea lappa* EO), α -humulene, and caryophyllene (from *Cordia*

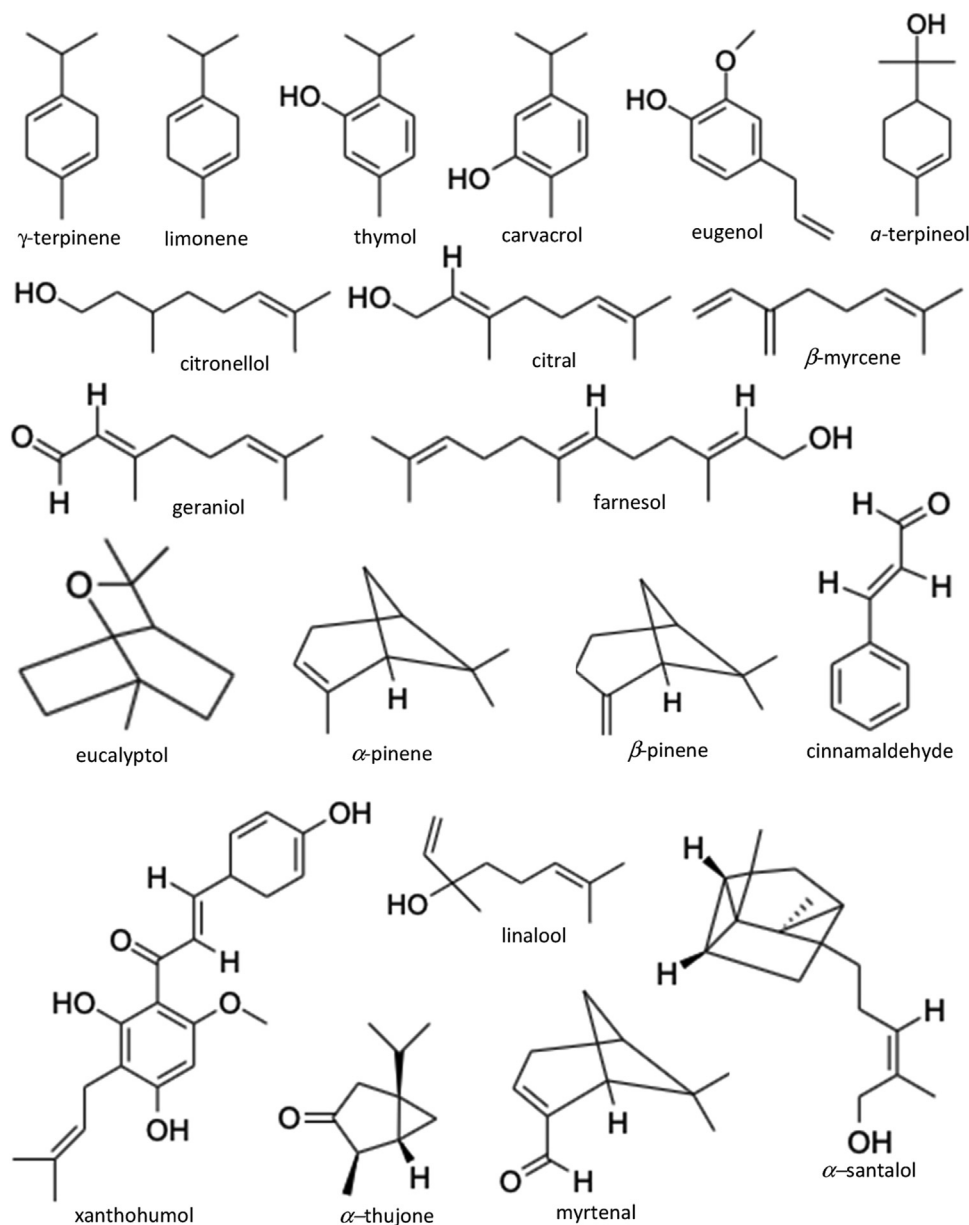


FIGURE 11.4 Chemical structures of common bioactive molecules isolated and identified from essential oils: anti-inflammatory (farnesol, limonene, citral, myrtenal), antibacterial (eucalyptol, α -thujone), antifungal (carvacrol, thymol, citronellol, citral, cinnamaldehyde, eugenol), antiviral (α -pinene, β -pinene, γ -terpinene, thymol, carvacrol, limonene), antiparasitic (thymol, eugenol, geraniol, carvacrol, limonene, eucalyptol, geraniol), antiplasmodial (α -pinene, β -myrcene, thymol, xanthohumol, α -santalol, α -thujone).

verbenacea EO) showed anti-inflammatory activity by inhibiting nuclear factor kappa B (NF- κ B) from inducing the expression of various pro-inflammatory genes responsible for encoding cytokines and chemokines [68]. Myeloperoxidase (MPO) activity is another example which EO components could inhibit or block. MPO is a pro-inflammatory enzyme stored inside azurophilic granules of neutrophilic granulocytes, and its activity generates reactive molecules which are associated with inflammations and inflammatory-related diseases [69]. Monoterpene oxide eucalyptol (1,8-cineole) can significantly reduce MPO activity and reduce colon inflammation in rats [70]. When tea tree oil (derived from *Melaleuca alternifolia*) was applied topically in mice, it reduces edema, which is attributed to the presence of terpinene-4-ol and α -terpineol [71]. Eugenol can also reduce edema in rats. Carvacrol and borneol are few more examples of anti-inflammatory components found to inhibit COX-2 enzyme and pro-inflammatory cytokine mRNA in mice [72].

11.5.2 Antimicrobial biomolecules

More than 1.5 million annual deaths worldwide result from invasive fungal infections alone [73]. EOs could be a potential source of new antimicrobial drug leads as they can disrupt cytoplasmic membrane or membrane proteins and subsequently causing cytoplasmic leakage, cell lysis, and eventually apoptosis [74]. Crude EOs from *Leonurus sibiricus* and *Origanum vulgare* can inhibit the growth of about 10 bacterial strains, including methicillin-resistant *Staphylococcus aureus* and vancomycin-resistant *Enterococcus* spp. [27,75]. Many compounds including, citronellol, octanal, decanal, nonanal, β -pinene, linalool, γ -terpinene, linalool, limonene, α -terpineol (from *Citrus reticulata* EO), carvacrol, thymol (from *Thymus vulgaris* EO), and citronellol and geraniol from *Rosa damascene* showed excellent antifungal

activities [76,77]. However, the exact mechanism of action of EO volatiles on fungi is still unclear. EOs can probably inhibit fungal sporulation, disrupt cell wall and cell membrane, and also interrupt the hyphal germination and elongation [78]. Among many other EO-yielding plants, *Citrus* species has attracted greater attention as a potential source of antimicrobials owing to high EO yield and good antimicrobial activity. More than 29 Citrus-derived compounds have shown antifungal activities, and Jing et al. [74] has extensively discussed the subject in their review article. EOs from other species such as *Ocimum gratissimum* (African basil), *Thymus vulgaris* (thyme) and *Cymbopogon citratus* (lemongrass) can inhibit conidial germination and the mycelial growth of three fungi (*Fusarium moniliforme*, *Aspergillus flavus*, *A. fumigatus*) [79]. However, unlike antibacterial activity, which is extensively studied, there are continued efforts toward enhancing the understanding of the mode of action of EOs on fungal spores and mycelial cells.

EOs have also shown promising activity against many pathogenic viruses including, SARS-CoV-2. Out of 171 EO components analyzed for molecular docking with SARS-CoV-2 main protease (SARS-CoV-2 M^{Pro}), (*E*)- β -farnesene showed best docking score suggesting that EOs may provide relief to COVID19 symptoms [80]. EO of *Laurus nobilis* also showed inhibitory activity on SARS-CoV-2 at IC₅₀ of 120 μ g/mL [81]. EOs also have inhibitory effects against herpes simplex virus (HSV-1 and HSV-2) and dengue virus. EO components including α -pinene, β -myrcene (from *Juniperus oxycedrus* ssp. *oxycedrus*) [81], γ -terpinene, terpinen-4-ol, and α -terpinene (from *Melaleuca alternifolia* EO) [82] are few examples showing activity against HSV-1 and HSV-2.

11.5.3 Antiparasitic and anthelmintic biomolecules

Parasitic - infection causes substantial health burden to more than 1.5 billion people [83].

For example, *Schistosoma mansoni* caused *schistosomiasis* to at least 230 million people in 2014 [84] and it is estimated that about 700 million are still at risk of its infection [85]. Plant EOs could be a potential source of antiparasitic drugs, particularly the terpene component of the EOs. A review by de Moraes [86] reported about 17 antischistosomal terpenes including (+)-limonene epoxide, rotundifolone, carvacryl acetate, and phytol. Godinho et al. [87] tested EO from *Tanacetum vulgare* against adult human blood fluke *S. mansoni* but was found inactive compared to its crude extract. Most of the anthelmintic activities of EOs are tested against ruminant nematode *Haemonchus contortus* [88]. More than 20 EO components are quite promising against various species of *Leishmania* [89], a lead cause of leishmaniasis, which is categorized as one of the neglected tropical diseases.

11.5.4 Antimalarial biomolecules

The global burden of malaria caused by *Plasmodium falciparum* and *P. vivax* has declined substantially since 2000, but infection with *P. falciparum* is still a frequent event in many countries [90]. Every year, about 100–400 million people are infected by *P. vivax* and *P. falciparum* [91,92]. Plants have served as a vital source of antimalarial drugs since the discovery of first antimalarial drug “quinine” from the bark of *Cinchona* species -in 1820 [93]. The decline in the efficacy of antimalarial drugs including chloroquine, to which *P. falciparum* has developed a resistance [94] demands alternative sources of new antimalarial drugs, and EOs has received considerable attention owing to its antiparasitic activities [95]. Many EOs and their components have shown exciting in-vitro antiparasmodial activities. For example, (*E*)-isoapiol isolated from the EO of Bhutanese medicinal plant *Pleurospermum amabile* showed significant *invitro* antiparasmodial activity against multidrug-resistant strains of *P. falciparum* (TM4/8.2 and

K1CB1) [96]. Other examples of EO components showing antiparasmodial activity against *P. falciparum* are thymol (isolated from *Satureja thymbra*) [97], hinokitiol (isolated from *Thujopsis dolabrata*) [98], and xanthohumol (isolated from *Humulus lupulus*) [99].

The elimination of malaria with current protocol is extremely challenging. Scientific evidence indicated that topical or spatial repellents could help prevent diseases like malaria [100]. Although there are few synthetic repellents such as DEET (*N,N*-diethyl-3-methylbenzamide), consumers are seeking natural-based repellents over synthetic products. Synthetic repellents cause toxic reactions and damages clothes [101]. Many studies are indicating the potential of EOs as insect repellents [102]. For instance, three commercially available *Melaleuca*-based formulations are effective at repelling mosquito *Aedes vigilax* [103]. EOs from Argentinian aromatic plants *Baccharis spartioides*, *Rosmarinus officinalis*, and *Aloysia citriodora* were found as an excellent repellent against *Aedes aegypti* at a concentration of 12.5% [101]. EOs from *Cymbopogon nardus*, *C. excavates*, *Eucalyptus maculata*, *Mentha piperata*, and *Azadirachta indica* also showed potential as mosquito repellent [104]. EO components thujone, eucalyptol, and linalool possess similar properties to the main ingredients of synthetic repellent, DEET [105].

11.5.5 Anticancer and antiproliferative biomolecules

EOs produced by medicinal and aromatic plants have gained renewed attention in the drug discovery research [106]. Studies have shown EOs as potential therapeutic choice against cancer [107]. Currently, there is not even single effective and noninvasive anticancer therapies, and it is going to become more challenging as diagnosed cancer cases are expected to increase from 13.3 million in 2010 to 20 million by 2030 [108]. EOs from medicinal and aromatic

plants offer fresh hope for potential anticancer drug lead(s). For instance, Chinese propolis could inhibit the proliferation of colorectal cancer cells [109] and similarly, *T. vulgaris* EO could also prevent the growth of human head and neck squamous cell carcinoma [110]. Many EO components are known to mediate cancer cells death via arresting the cancer cell proliferation and metastasis. γ -humulene, α -elemene, β -caryophyllene oxide, α -santalol, α -humulene, thymol, nimbolide, geraniol, eugenol, carvacrol, azadirachtin, and nimbolide are few examples reported to have potent anticancer activity [111]. Cancer patients undergoing chemotherapeutic treatments experience depression and anxiety, and EOs has proved effective in easing these symptoms [112].

11.6 Trade and commercial applications of essential oils in healthcare

The global EOs market demand is estimated to increase by 7.5% between 2020 and 2027 with increasing demand from food and beverages, personal care and cosmetics, and aromatherapies [11]. Multiple health benefits of EOs is projected to drive a similar demand in the pharmaceutical and medical applications. Unlike other conventional drugs and medicines, comparatively lesser side effects of the EOs is a major factor for its market growth. India, China, Brazil, Egypt, Mexico, United States, and Canada are major EO-producing countries, while Japan, United States, and European Union including France, Germany, and the United Kingdom are biggest importers [113]. Of the total \$6.22 billion export value in 2018, India is the largest contributor with 13.2% followed by United States (12.6%), China (9.08%), France (8.26%), and Brazil (7.13%). There are many examples of commercial EOs, and some of countries and type of EOs they produce are given in Fig. 11.1. More than 32 countries produce 50 commercial EOs and

approximately 50% of them are cultivated [113]. Lemon, orange, peppermint, lime, citronella, corn mint, spearmint, cloverleaf, geranium, jasmine, eucalyptus, rosemary, lavender, and tea tree oils are most commonly marketed EOs [11,113].

EOs are mostly used in relaxation therapies (aromatherapy, spa, flavorings, and fragrances), food and beverages, cosmetics, and toiletries. Among relaxation therapies, aromatherapy is considered as one of the safest and cheaper treatment options for both acute and chronic stress. When EO is inhaled, it binds to receptors and passes the signal to limbic and hypothalamus parts of the brain through the olfactory bulb, and subsequently release neurohormones like serotonin and endorphin which are responsible for producing a feeling of relief or relaxation [3]. Patient's stress level is directly associated with his/or her chances of recovery, and stress biomarker level is almost always significantly lower in survivors than in nonsurvivors. Patients inhaling a relaxing EO before various medical procedures such as dental surgery, radiotherapy, colonoscopy, and prolonged renal hemodialysis help them reduce their stress level. Pleasant smell of EOs from sweet orange (*Citrus sinensis*) and lavender (*Lavandula angustifolia*) subside anxiety and considerably improve mood in patients before their dental treatment [114]. Frankincense, mandarin, rose, and bergamot EOs also possess the potential to reduce stress level.

Chronic stress often leads to chronic illness which is associated with depression, reduced cognition, and increased pain sensitivity. Pain resulting from prolonged treatment and a lengthy stay in acute care units is a global issue, and one of the common symptoms in a clinical setting. Many countries, including the United States, Canada, Australia, and Japan, are bearing considerable costs in the management of chronic pain. Every year, the United States spends over \$365 billion, Canada over \$6 billion, and Australia over \$34.4 billion [115,116]. Inhaling aromatic EOs has been

found to alleviate chronic pain and enhance tolerance to pain. Cardamom, lavender, and geranium EOs are few examples which can increase the adsorption as well as the potency of pain medicines. The pain-alleviating potency of EOs is mainly attributed to analgesic properties of EOs or their chemical constituents. Many EOs and their isolated components showed analgesic or antinociceptive activity in animal models of rats and mice. Few examples of potent analgesic compounds are (-)-linalool (major constituent in lavender and clary sage EOs) and myrcene from *C. citratus* EO [117,118]. There are clinical trials conducted on aromatherapy for the treatment of stress and anxiety in patients with critical illnesses like cancer. Aromatherapy using EOs from lavender and peppermint can improve the sleep quality in cancer patients [119]. A mixture of EOs from thyme, rosemary, lavender, and cedarwood proved effective against alopecia areata when applied into scalp daily [120]. Thus, aromatherapy and massage with EOs are integrated into pain management programs in several countries, including the United Kingdom, United States, Canada, and Australia.

EOs also possess excellent antimicrobial property, and the demand for the use of EOs from herbs and species as preservatives in food and beverages is escalating since consumers prefer natural preservatives over synthetic [121]. EO components can prevent microbial growth in the food matrix and increases the shelf-life of food commodities. There are numerous examples of EOs as a potential food preservative: citral, citronella, citronellol, eugenol, farnesol, and nerol prevents fungal infections in chili seeds and fruits for at least upto 6 months [122] (Table 11.2). Crude EOs from *Cymbopogon flexuosus*, *C. nardus*, and *Ocimum basilium* can keep banana afresh upto 21 days by preventing a fungal infection, anthracnose [162]. Groundnuts fumigated with *Putranjiva roxburghii* EO can prevent infection from *Aspergillus flavus*, and *A. niger* upto 6–12 months [163]. *C. flexuosus* EO can also

protect *Malus pumilo* fruits from rotting upto 3 weeks when applied at a concentration of 20 $\mu\text{L/mL}$ [164]. The preservative effects of EOs varies depending upon the plant source and mode of applications. Besides the application of EOs in food preservation, they are also used in the development of edible films, and encapsulations and active packaging.

11.7 Risks and dangers of essential oils

EOs contain a number of chemical components which may sometimes undergo further metabolism and as a result produce potent toxicants [165]. Due to the complex mixture compounds, it is challenging to sort out which chemical is either safe or unsafe for human consumption. The United States Food and Drug Administration (FDA) have categorized EOs as “generally safe products” and many EOs are approved as food additives [166]. However, when an EO is applied in excess in the food matrix, it may end up having an unacceptable organoleptic impact. Most of the poisonings result from accidental exposures, therapeutic errors, mistaking EOs for other non-toxic liquid pharmaceuticals like cough liquids, and also due to misinformed misuses [167].

People who have previous records of asthma and allergies are more sensitive to chemicals present in EOs, and they are likely to develop adverse conditions including dermatitis and bronchospasm, even in contact with low therapeutic dosage [165]. Infants and children are at higher risks as their liver may not be able to fully metabolize EOs components due to immature hepatic enzymes. A study conducted by Lee et al. [167] on EO exposure poisoning in Australia found that as per the New South Wales Poisons Information Center, there is an increase of 5.3% cases every year and more than half involved children. EOs may cause irritation of eyes, mucous membrane, and skin, and the photosensitization by those oils containing

TABLE 11.2 Commercially available essential oils and their reported health benefits.

Commercially available essential oil	Plant source/ingredients	Chemical constituents	Country	General uses	Biological activities
Arborvitae	<i>Thuja plicata</i>	Methyl thujate	Canada	Promote healthy skin when applied topically.	Anti-inflammatory [123]
Bergamot	<i>Citrus bergamia</i>	Limonene, linalyl acetate	Italy	Produce calming aroma and used for massage therapy; promote healthy skin.	Wound healing [124]; Anti-tumor [125]; Neuroprotective [126]
Cedarwood	<i>Juniperus virginiana</i>	α -Cedrene, cedrol, thujopsene	United States of America	Produce relaxing aroma; promotes skin health.	Antibacterial [127]; Larvicidal [128]
Cinnamon	<i>Cinnamomum zeylanicum</i>	Cinnamaldehyde, eugenol	Sri Lanka, Madagascar	Produces stimulating and invigorating aroma when diffused; flavoring agent.	Antibacterial [129]
Clove	<i>Eugenia caryophyllus</i>	Eugenol	Madagascar	Used in massage therapy.	Antimicrobial (https://www.doterra.com)
Cypress	<i>Cupressus sempervirens</i> , <i>C. lusitanica</i>	α -Pinene, 3-carene, limonene	Spain	Produces uplifting aroma; improves skin appearance.	Antimicrobial, antibiofilm activities [130]
Douglas fir	<i>Pseudotsuga menzeisii</i>	β -Pinene, α -pinene, 3-carvene, sabinene	New Zealand	Clearing airways when inhaled and provides uplifting aroma.	Antiproliferative [131]
Eucalyptus	<i>Eucalyptus cneorifolia</i>	α -Terpineol, eucalyptol.	Australia, Kenya, South Africa	Used for cuts, burns, cough and cold, pain, acne, insect bites and abrasions. Also used as antiseptic, insect repellent, mouthwash, hand and skin cleaner.	Antibacterial and antifungal (https://www.doterra.com)
Frankincense	<i>Boswellia carterii</i>	Fractionated coconut oil, Frankincense resin	Ethiopia, Somalia, Oman	Used for relaxation; improves skin glow.	Antimicrobial [132]; Anti-inflammatory [133]
Helichrysum	<i>Helichrysum italicum</i>	Neryl acetate, α -pinene, γ -curcumene	Greece, Vatican City	Skin care and massage therapy oil.	Anti-oxidant [134]; Antimicrobial [135]
Lavender	<i>Lavandula angustifolia</i>	-	Bulgaria	Soften and improve skin appearance; calming and relaxing.	Antibacterial [136]
Lemon	<i>Citrus limon</i>	Limonene, β -pinenes, γ -terpinene	Italy, Kenya	Refreshing and uplifting effects.	Anti-oxidant [137]; Anti-obesity [138]; Anti-anxiety [139]; Analgesic [140]; Neuroprotective [141]; Antifungal [142]
Lemongrass	<i>Cymbopogon</i> spp.	Geranial, neral	India, Bhutan	Produces refreshing aroma; flavoring food.	Antimicrobial [143]
Magnolia	<i>Michelia alba</i>	Linalool	China	Gives calm and relaxing feeling; promote healthy skin.	Insecticidal [144]; Antimicrobial [145].
Manuka	<i>Leptospermum scoparium</i>	Leptospermone, (E)-calamenene, α -pinene, cadina-3, 5-diene	New Zealand	Promotes relaxing environment when diffused; produce refreshing aroma; promotes the appearance of smooth and healthy skin.	Antibacterial [146]; Anti-inflammatory [147].

Marjoram	<i>Origanum majorana</i>	Terpinene-4-ol, sabinene hydrate, γ -terpinene	Egypt	Gives calming and uplifting aroma; food flavor.	Antimicrobial [148]; Anti-oxidant [149]; Anticancer [150]
Mentofin	Eucalyptus and peppermint	Eucalyptus oil, menthol, liquid builders, saponins	Germany	Helps to prevent respiratory problems due to post vaccination reactions.	Antiviral [151]
Neroli	<i>Citru aurantium</i>	Linalool, linalyl acetate, limonene.	Egypt	Gives calming and uplifting aroma and gives relaxation.	Antimicrobial and antioxidant [152]
On Gaurd	Wild orange peel, Clove bud, Cinnamon leaf, Cinnamon bark, Eucalyptus leaf, Rosemary leaf/flower EOs	Wild orange peel, clove bud, Cinnamon leaf, Cinnamon bark, Eucalyptus leaf, Rosemary leaf/flower EOs	United States	Food flavor, topical application, cleaning surfaces.	Antiviral [153]
Oregano	<i>Origanum acutidens</i> , <i>O. bilgeri</i>	Carvacrol, thymol, γ -terpinene, <i>p</i> -cymene, <i>trans</i> - β -caryophyllene.	Turkey	Cleansing agent; help reduce the appearance of blemishes.	Anti-inflammatory, antioxidant, antiproliferative, antimicrobial [154]
Peppermint	<i>Mentha piperita</i>	Menthol, menthone, eucalyptol	United States	Improve cognitive; reduce/relieve flatulence, stress, and headache; used in aromatherapy to relieve cough.	Antibacterial [155]
Patchouli	<i>Pogostemon cabin</i>	α -Bulnesene, α -guaiene, patchouli alcohol	Indonesia	Improves skin complexion and reduces appearance of wrinkles; gives balancing aromatic effects on emotions.	Anti-oxidant [156]; Antibacterial [157]
Sandalwood	Wood of <i>Santalum album</i>	<i>cis</i> - β -Santalol, <i>cis</i> - α -santalol	Australia, Hawaii island	Topical application as a skin serum or massage oil. Facial application (recommended 1% dilution).	Anti-inflammatory, antiseptic, aphrodisiac, and effective relaxant (https://www.doterra.com).
Tangerine	<i>Citrus reticulata</i>	Limonene	Brazil	Energizing and refreshing effects.	Antimicrobial [158]
Tea tree	<i>Melaleuca alternifolia</i> and <i>M. linariifolia</i>	Terpinen-4-ol, γ -terpinene, α -terpinene, 1,8-cineole, terpinolene, <i>p</i> -cymene, α -pinene, α -terpineol.	Australia, Kenya	Cleansing skin and nails; promotes skin complexion.	Antiviral [159]
Vetiver	<i>Vetiveria zizanioides</i>	Carvone, isovalencenol, khusimol	Dominican Republic	Calming effects; used as bathing and massage oil for relaxing effects.	Antiproliferative activity [160]
Ylang-ylang	<i>Cananga odorata</i>	Germacrene, caryophyllene, farnesene	Madagascar	Improves appearance of skin and hair (topical application); gives calming and uplifting aroma.	Antimicrobial [161]

Note: Information on general uses of commercial essential oils are mostly retrieved from the <https://www.doterra.com>.

aldehydes and phenols are most common. For example, *Citrus bergamia* EO, which has furocoumarins, causes phototoxicity [3]. Contact sensitization caused by some EOs is mainly attributed to oxidation of monoterpenes due to improper storage [168]. Toxic EOs can affect the erythrocytes morphology leading to increased level of total erythrocyte content (TEC) after exposure (externally or internally). EOs from *Murraya koenigii*, *Skimmia laureola*, and *Boenninghausenia albi-flora* caused an increased level of TEC and sodium ions in the blood of Wistar rats [169]. Odeyemi et al. [170] also found similar effects in their study involving EOs. Thus, some EOs may not be safe to consume or apply topically. Studies have also shown that EOs may cause latent toxicities, such as cancer and congenital malformations. For example, mice exhibit teratogenic effects when treated with *Plectanthus fruticosus* EO [165]. Many studies highlight possible poisoning that may result from accidental high dose or raw consumptions of EOs or plant parts rich in EOs. For instance, eating a few grams of EO-rich nutmeg (e.g., seed kernel of *Myristica fragrans*) may cause liver failure and death [171]. Eucalyptol (which is a major component of eucalyptus oil) can impair the hepatic microsomal enzyme system involved in the normal metabolism of toxic drugs and chemicals and can cause neurotoxicity [165]. Another example is pennyroyal oil distilled from *Hedeoma pulegiodes* and *Mentha pulegium* with an aromatic ketone “pulegone” as the main component. Pulegone upon bioactivation produces its chemical derivative “menthofuran,” both of which depletes the glutathione level in the cells; subsequently, cells are easily attacked by oxidizing free radicals [172]. Peppermint (*Mentha* spp.), tea tree, lavender, clove, and mugwort (*Artemisia* spp.) poisonings are few more examples [167]. Activated charcoal and N-acetylcysteine (NAC) are often tried orally to overcome EOs poisoning [165,173] but all these demands more scientific testing in animals as well as human models for better decision-making.

11.8 Conclusion

EOs are a highly complex mixture of aromatic and lipophilic volatile compounds, and monoterpenes constitute 90% of EOs. So far, more than 3000 EOs have been extracted and analyzed from about 17,500 plants, mostly from angiosperm families. Brazil, China, Egypt, India, Mexico, Guatemala, and Indonesia are the major EO-producing countries, where Brazil alone produces more than 40,000 tonnes of EO annually; while Japan, United States, and Western Europe are major consumers. The yield and quality of EOs are influenced by multiple factors including morphogenetic, parts used for extraction, harvesting seasons, postharvest storage conditions and method of extractions. EOs have been used historically for treating many diseases due to their interesting biological activities. The EO has a wide range of pharmacological properties such as antimicrobial, antioxidant, anti-inflammatory, antiproliferative or anticancer, neuroprotective, antimalarial, and antidiabetic activities. Due to their broad-spectrum pharmacological activities and particularly, antimicrobial activity, more than 300 EOs are available in the food and fragrance markets, and their numbers are soaring as consumers prefer natural preservatives over synthetically processed foods. However, EOs are a complex mixture of a number of compounds, which is challenging to sort out toxic from useful compounds. Poisonings from accidental consumptions of EOs are still common, particularly among children and pregnant women. A considerable amount of studies has shown the potential scope of EOs from plants in the pharmaceutical industry for developing therapeutics against various diseases. However, future endeavors should focus on isolating individual bioactive components from EOs followed by rigorous clinical studies using animal models and human volunteers to translate them into final palatable therapeutic drugs.

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